Discussion 2 Notes

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October 06, 2026

§I Warm-Up

> (Warm-Up) Order the following numbers from least to greatest:
>
> $$\sqrt{2027}, \quad 2027^2, \quad 2^{2027}, \quad 2027^{2027}, \quad 2027!, \quad \ln 2027, \quad 3^{2027}.$$

Solution. The correct order is

$$\ln 2027 < \sqrt{2027} < 2027^2 < 2^{2027} < 3^{2027} < 2027! < 2027^{2027}.$$

Some of my intuition:

- $\ln 2027 < \sqrt{2027}$ because $\ln 2027$ is certainly less than 40 (2027 is less than e^{40}) and 40 is less than $\sqrt{2027}$. Or make the plot of $\ln x$ and $\sqrt{x}$ and we will see that $\ln x$ is always below $\sqrt{x}$.
- $\sqrt{2027} < 2027^2$ because $\sqrt{2027}$ is less than 2027 and 2027 is less than 2027^2.
- $2027^2 < 2^{2027}$ because exponential grows very fast, $2^{10} = 1024$.
- $2^{2027} < 3^{2027}$ because $2 < 3$.
- $3^{2027} < 2027!$ because they are both products of 2027 numbers, but 3^{2027} is the product of 2027 threes, while 2027! is the product of $1, 2, 3, \ldots, 2027$, which is much larger.
- $2027! < 2027^{2027}$ because 2027^{2027} is the product of 2027 copies of 2027, which is even larger.

§II Limit Comparison Test

Consider the series

$$\sum_{n=1}^{\infty} \frac{n - \ln n}{n^2 + \sqrt{n} + 1}.$$

Intuitively, for very large n, we want to say that n is so large compared to $\ln n$ that the numerator is effectively n, and similarly, n^2 is so large compared to $\sqrt{n}$ and 1 that the denominator is effectively n^2. Therefore, we are just looking at the series

$$\sum_{n=1}^{\infty} \frac{n}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n}$$

which is a divergent p-series with $p = 1$.

The **limit comparison test** is what makes this intuition rigorous. Let us introduce the notation that

$$a_n \ll b_n \quad \text{if} \quad \lim_{n\to\infty} \frac{a_n}{b_n} = 0 \quad \text{or equivalently} \quad \lim_{n\to\infty} \frac{b_n}{a_n} = \infty.$$

The limit comparison test states that

> If $a_n \geq 0$ and $b_n \geq 0$ are nonnegative terms, then
>
> 1. If $\lim\limits_{n\to\infty} \dfrac{a_n}{b_n} = c$ and $0 < c < \infty$, then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.
> 2. If $a_n \ll b_n$, then the convergence of $\sum b_n$ implies the convergence of $\sum a_n$ and the divergence of $\sum a_n$ implies the divergence of $\sum b_n$. It is as if

$$\sum a_n \leq \sum b_n$$

except that technically, this inequality is only true if the we restrict the lower bounds of the series to be sufficiently large.

If we return to our example, we have concluded that we should compare the series to $\sum_{n=1}^{\infty} \frac{1}{n}$ and this is exactly what we will do. Rather than using nested fractions, it is the most elegant to multiply $\frac{n-\ln n}{n^2+\sqrt{n}+1}$ by the inverse of $\frac{1}{n}$, so we have

$$\begin{aligned}\lim_{n\to\infty} \frac{n-\ln n}{n^2+\sqrt{n}+1} \cdot \frac{n}{1} &= \lim_{n\to\infty} \frac{n^2 - n\ln n}{n^2+\sqrt{n}+1} \\ &= \lim_{n\to\infty} \frac{1 - \frac{\ln n}{n}}{1 + \frac{1}{\sqrt{n}} + \frac{1}{n^2}} \\ &= 1\end{aligned}$$

where we used the fact that

$$\lim_{n\to\infty} \frac{\ln n}{n} \overset{\text{LH}}{=} \lim_{n\to\infty} \frac{\frac{1}{n}}{1} = 0.$$

Before we move on to exercises, let us note that

$$\text{constant} \ll \ln n \ll \cdots \ll n^{\frac{1}{3}} \ll n^{\frac{1}{2}} \ll n \ll n^2 \ll n^3 \ll \cdots \ll e^n \ll n! \ll n^n.$$

Remembering this will help us identify the fastest growing terms in the numerator and denominator of a series.

(Exercise) Determine whether the following series converge or diverge.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{2^n+5}{3^n}$
2. $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$
3. $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{e^n}$ (hint: split $\displaystyle\frac{n^2}{e^n} = \frac{n^2}{e^{n/2}} \cdot \frac{1}{e^{n/2}}$)

Solution.

1. The fastest growing term in the numerator is 2^n and the fastest growing term in the denominator is 3^n, so we will compare to $\sum_{n=1}^{\infty} \frac{2^n}{3^n}$. We have

$$\lim_{n\to\infty} \frac{2^n+5}{3^n} \cdot \frac{3^n}{2^n} = \lim_{n\to\infty} \left(1 + \frac{5}{2^n}\right) = 1.$$

Since $\sum_{n=1}^{\infty} \frac{2^n}{3^n}$ is a convergent geometric series, we conclude that $\sum_{n=1}^{\infty} \frac{2^n+5}{3^n}$ converges.

This problem can be more easily solved by splitting the series into two parts. This is mistake in problem design on my part. A better problem is perhaps

$$\sum_{n=1}^{\infty} \frac{2^n+5}{3^n-2^n}.$$

2. For large n, 1 is negligible compared to n^2, so we will compare to $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2}} = \sum_{n=2}^{\infty} \frac{1}{n^2}$. We have

$$\lim_{n\to\infty} \frac{1}{n\sqrt{n^2-1}} \cdot n^2 = \lim_{n\to\infty} \frac{n}{\sqrt{n^2-1}} = \lim_{n\to\infty} \frac{1}{\sqrt{1-\frac{1}{n^2}}} = 1.$$

   Since $\sum_{n=2}^{\infty} \frac{1}{n^2}$ is a convergent p-series with $p = 2 > 1$, we conclude that $\sum_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$ converges.
3. We expect that $n^2 \ll e^{n/2}$, so we will expect

$$\frac{n^2}{e^n} = \frac{n^2}{e^{n/2}} \cdot \frac{1}{e^{n/2}} \ll \frac{1}{e^{n/2}} = \left(\frac{1}{e^{1/2}}\right)^n.$$

   Explicitly, we need to write

$$\lim_{n\to\infty} \frac{n^2}{e^n} \cdot \frac{e^{n/2}}{1} = \lim_{n\to\infty} \frac{n^2}{e^{n/2}} \overset{\text{LH}}{=} \lim_{n\to\infty} 4\frac{n}{e^{n/2}} \overset{\text{LH}}{=} \lim_{n\to\infty} \frac{8}{e^{n/2}} = 0.$$

   Since $\sum_{n=1}^{\infty} \frac{1}{e^{n/2}}$ is a convergent geometric series ($|r| = e^{-1/2} < 1$), we conclude that $\sum_{n=1}^{\infty} \frac{n^2}{e^n}$ converges.

§III Ratio and Root Tests

The ratio and root tests extend our ability to determine convergence and divergence of series that involve products of factorial terms $n!$ and power terms like n^k, a^n, and n^n.

For series with factorials, the ratio test is essentially the exclusive test. For example, consider

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}.$$

We will set $a_n = \frac{2^n}{n!}$ and decide whether

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

is less than 1 (convergent), greater than 1 (divergent), or equal to 1 (inconclusive). The elegant way to write a_{n+1}/a_n is not through nested fractions but rather multiplying a_{n+1} by the inverse of a_n:

$$a_{n+1}\frac{1}{a_n} = \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} = \frac{2}{n+1}.$$

Thus, we have

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{2}{n+1} = 0 < 1,$$

which implies that the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges.

The root test is effective when the terms are products of power terms. For example, consider

$$\sum_{n=1}^{\infty} \frac{n^7}{n^n}.$$

We can apply the root test by looking at the limit

$$\lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^{7/n}}{n} = \lim_{n\to\infty} \frac{\left(n^{1/n}\right)^7}{n} = 0 < 1.$$

Similar to the ratio test, a limit of less than 1 implies convergence, a limit greater than 1 implies divergence, and a limit equal to 1 is inconclusive.

An identity that is worth remembering for the root test is that

$$\lim_{n\to\infty} n^{1/n} = 1,$$

which we used in the previous example.

If you want to join the root test fan club, you need to know that

$$\lim_{n\to\infty} (\text{polynomial in } n)^{\frac{1}{n}} = 1$$

and you need know that the trick to show this is to apply the identity $x = e^{\ln x}$ and its byproduct $x^y = e^{y \ln x}$, so we have

$$\lim_{n\to\infty} (\text{polynomial in } n)^{\frac{1}{n}} = \lim_{n\to\infty} e^{\frac{1}{n}\ln(\text{polynomial in } n)}.$$

From here, we can apply L'Hôpital's rule to show that the limit is 1.

Try this for $n^{\frac{1}{n}}$ and for $(n^2+1)^{\frac{1}{n}}$.

(Exercise) Determine whether the following series converge or diverge.

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(2n)!}{n+2}$
2. $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{e^{n^2}}$

Solution.

1. We will apply the ratio test. We calculate

$$\begin{aligned}
\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| &= \lim_{n\to\infty} \frac{(2(n+1))!}{n+3}\cdot\frac{n+2}{(2n)!} \\
&= \lim_{n\to\infty} \frac{(2n+2)!}{n+3}\cdot\frac{n+2}{(2n)!} \\
&= \lim_{n\to\infty} (2n+2)(2n+1)\frac{n+2}{n+3} \\
&= \lim_{n\to\infty} (2n+2)(2n+1)\frac{1+2/n}{1+3/n} \\
&= \infty > 1.
\end{aligned}$$

   Since the limit is greater than 1, we conclude that the series $\sum_{n=1}^{\infty} \frac{(2n)!}{n+2}$ diverges.

2. Ratio test also works well, but we will apply the root test. We calculate

$$\lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \left(\frac{2n}{e^{n^2}}\right)^{\frac{1}{n}} = \lim_{n\to\infty} \frac{2^{1/n} n^{1/n}}{e^n} = 0 < 1.$$

   Since the limit is less than 1, we conclude that the series $\sum_{n=1}^{\infty} \frac{2n}{e^{n^2}}$ converges.